

ANIRUDH MAKULURI

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SUMMARY

Software engineer who builds and operates production systems end to end across backend, infrastructure, and user-facing products. Scaled a real-time platform to 12K+ users and built SmartDeploy, a system that analyzes codebases and executes cloud deployments with automated recovery and live system feedback. Recently focused on building LLM-driven systems that take actions on top of real infrastructure.

EDUCATION

M.S Computer Science <i>Arizona State University, Tempe, AZ (ASU)</i>	August 2024 – May 2026 <i>GPA: 4.00/4.00</i>
B.E. Electronics & Instrumentation <i>Birla Institute of Technology and Science Pilani (BITS Pilani)</i>	August 2018 – May 2022

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, C/C++, Go, SQL, Java
Frontend: React, React Native, Next.js, Zustand, TanStack Query, Tailwind CSS, Redux, HTML
Backend/APIs: Node.js/Express, FastAPI, Socket.IO/WebRTC, Microservices
Databases: PostgreSQL/Supabase, Neo4j, Firestore, MongoDB, DynamoDB
Cloud/DevOps: AWS (EC2, SSM, ECR, ALB, ECS), GCP, Docker, GitHub Actions CI/CD, Terraform, Kubernetes
AI/ML: TensorFlow, PyTorch, RAG/sentence-transformers, Scikit-Learn, NLP/Knowledge Graphs, LangChain, LangGraph

PROFESSIONAL EXPERIENCE

Shardings — Software Engineer (Early Engineering Team) - Designed and implemented scalable architecture for a full-stack video conferencing platform serving 12K+ users, ensuring reliable cross-platform feature delivery across web and mobile. - Integrated Google Cloud Speech-to-Text and ChatGPT API to automatically transcribe and summarize live standups, eliminating 5 hours/week of manual note-taking per team and making recaps searchable for later audits. - Improved system performance by integrating Google Cloud CDN and optimizing asset delivery, reducing median load latency from 3 seconds to 800 ms and improving reliability of media-heavy workflows. Owned and operated Node.js backend services powering 20+ REST APIs and 30+ WebSocket streams with 99.5% uptime, debugging production issues and optimizing for sub-200ms latency. - Implemented blue-green deployments for real-time media servers, enabling zero-downtime releases and seamless cross-version traffic migration for live user sessions.	June 2022 – August 2023
Shardings — Software Engineer Intern Implemented full payment gateway with Razorpay and Stripe integrations supporting 3 pricing tiers (Basic, Standard, Pro); designed secure webhook handling, idempotency, and error recovery, handling 100+ transactions per month with 99% success rate.	July 2021 – December 2021

PROJECTS

SmartDeploy — Preview-Driven Deployment Platform - Built a deployment platform (Next.js, Postgres, AWS, GCP) that analyzes GitHub repos, generates infrastructure (Docker, docker-compose, Nginx), enables pre-deployment previews and deploys to cloud services, supporting 70+ deployments. - Provisioned SmartDeploy on AWS EC2 using Terraform, and built the app deployment pipeline using CodeBuild, ECR, EC2, and ALB for reliable build, image push, and routing. - Implemented automated failure detection and recovery (health checks, retries, rollback) using deployment logs, achieving an 88% deployment success rate and reducing manual debugging effort. - Designed real-time deployment streaming (SSE) for logs, build steps, and service state, reducing dashboard latency from 5s to 1s using TanStack Query caching and GraphQL.	July 2025 – April 2026
SD-Artifacts — Infrastructure Generation Engine - Built an LLM-powered system (FastAPI, LangGraph, AWS Bedrock) to infer application architecture from GitHub repositories and generate production-ready Docker, docker-compose, and Nginx configurations. - Achieved 96% service-level F1 and 100% stack detection accuracy on 18 real-world JS/TS/Python repositories; reduced redundant analysis by 90% via caching with Supabase.	January 2026 – March 2026
Chatify — E2EE AI Chat Platform - Built a real-time chat system (React Native, React.js, Node.js, Socket.IO) with sub-500ms latency across mobile and web, integrating an AI assistant with semantic retrieval for context-aware conversations. - Implemented offline-first messaging (queued writes, retries, conflict handling) and end-to-end encryption (libsodium), ensuring consistent message delivery and ordering under unstable network conditions.	Dec 2024 – Jan 2026

ACHIEVEMENTS & CERTIFICATIONS

1st Place (Auth0 AI Agents) & 2nd Place (Google Track) — Innovation Hacks 2.0, ASU (600+ participants), April 2026 — Built <i>TravelMate</i>, AI travel companion with real-time voice call interface for itinerary assistance Certifications: Oracle Generative AI Professional; ML Specialization (Stanford/DeepLearning.AI)
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